



Seagate's new FreeAgent series of USB drives get new giant, a 640 GB version replaces the 500 GB at the top of the lineup

## Suggestive Google?

Google Suggest disabled in China due to porn

will look completely at home with your Apple Macbook and iPhone. There is a single large LED that is a capable diagnostics tool, it can remain steady, blink slower or faster and has three colour options – orange, green and blue that represent the status of the device. The LAN ports at the rear also have tiny LEDs, but there is no indication at the front. The software interface is very different from most routers and is available after installation of software present on CD. There are fewer options when compared to the firmware of other routers. The presence of a USB port means it can be hooked up to a printer or a USB drive. The Senao EnGenius ESR-9750G is really compact and well built with a glossy white body. The blue front LEDs are painfully bright and there are fewer options available in the



Belkin Wireless G Router

firmware menu although to be honest this will be a relief for newbies as the menu is really intuitive to work with. A hardware button is provided for enabling WPS, but there's no indicator for whether the feature is active or not. The Dlink DIR-655 is a classy looking router finished in a glossy white with a black band running around three sides. Build quality is exemplary and it has three large, detachable, external antennas. The front LEDs are bright and blue – too garish. Instead of cutting the shape of numbers into the plastic, Dlink chose to simply cut little squares, so the wired LAN indicators are bright blue little squares – this

felt slightly cheap and a letdown from the overall quality. The menu system is simple with a lot of options available. This router also has a USB port and can be used as a multimedia server. Netgear's WNR3500 was the largest router physically and it cannot be wall-mounted, thankfully it has a neat stand so takes up little space. It's built well and looks classy with a mixture of plain and glossy black finishes. A strip of blue LEDs flash along the top of the router and this becomes annoying because we could find no way to switch them off. After dark, it's super annoying and the blue colour hurts the eyes. This router has a very neat firmware interface that is shared by all Netgear routers – simple, well laid-out and pretty intuitive. The wireless repeating function is present. Content filtering is powerful and easy to set up with very transparent settings. Sites can be blocked on the basis of keywords, while sites can be blocked to certain or all IPs and all these activities can also be set on a toggle timer – useful if kids surf the internet at particular times or otherwise as a security feature. We noticed that Netgear's firmware interface is simpler than Linksys' – less jargon used, which is newbie friendly.

The WNDR3300 was the next Netgear product we looked at. It's large, but neat looking with a glossy black finish and a transparent bezel running right around the sides. There is a large, translucent dome atop the unit that doubles as a WPS button, the bright blue LEDs inside are overdone though and a part of the blue PCB used is visible through this button. There are two activity indicators for Wi-Fi – one each for 2.4 GHz and 5GHz bands. This is a dual simultaneous band router, meaning that not only can it create two wireless signals of two different frequencies, but it can do so simultaneously. In

## HOW WE TESTED

We divided the products into three categories:

1. Access Points
2. Routers (subdivided into two parts on the basis of price)
3. ADSL Routers.

In terms of features, we logged everything from the connectivity options to the firmware interface, build quality, security features and configurable settings. We did not give scores for all the features though, only the ones we felt were pertinent enough to warrant attention. The weights assigned to each feature were given on the basis of its relative importance in making the product more usable, functional or feature rich. Testing was done simulating all possible real-world conditions, for which we created the test 'zones' that we've described below. Please note that distances aren't as important for benchmarking any sort of 802.11 networks it's the obstructions between the emitting device and the recipient that matter. Metal along with brick and concrete cause the most signal attenuation. Even wooden furniture and an unclear line of sight (LOS) will scramble Wi-Fi signals. A reduction in signal strength brings about a crippling reduction to bandwidth and therefore slows data throughput. With this in mind, we designed each of these zones to actually take each of these wireless devices to their performance limits.

**Zone 1** – The laptop was placed within eight feet of the wireless device although a clear LOS was not allowed since the device was placed on a table, away from the edge

**Zone 2** – We placed the laptop within 30 feet of the router. This time we ensured that there was a good deal of obstructions. There was a concrete wall and also a couple of wooden partitions

**Zone 3** – Our test laptop

was placed 25 feet away from the router. Here we actually moved the laptop one floor lower. We also ensured the laptop wasn't exactly above the router placed a level below (this would be too easy). This was the most difficult test for the any wireless device as there are multiple obstructions and also a lot of metal in the ceiling structure that really messes with Wi-Fi signals.

Our first test involved a sequential transfer (one large file) and a random transfer (multiple small files of varying sizes) from a desktop PC wired via Ethernet to the wireless device and a laptop. The speeds were recorded in Megabytes per second (MBps). Please note that most manufacturers quote figures in Megabits per second (Mbps) where 1 byte = 8 bits.

Next we used PassMark's Performance Test as a benchmark. The inbuilt network benchmark simulates a data transfer and measures throughput in megabits per second (Mbps). PassMark's Wireless Mon 3.0 was used to measure signal strength as a percentage.

Next we streamed a 1080p video clip from the desktop PCs hard drive to the notebook wirelessly. We awarded points based on the entire viewing experience. We also streamed a DVD and watched for jerkiness in both cases. All this testing was repeated through the zones.

**Test PC:** Core 2 Duo E6700 (2.66 GHz), 2 GB DDR2 800 MHz RAM, Gigabyte P965 DQ6 motherboard, XFX GeForce 9600GT graphics card, 320 GB, 7200 rpm HDD, Intel 802.11n WiFi solution (Centrino 2)

**Test notebook:** Dell Studio XPS 16 – Core 2 Duo 2.53 GHz, 4 GB RAM, ATI Radeon 3670 graphics card, 320 GB, 7200 rpm HDD, Intel 802.11n WiFi solution (Centrino 2)

theory, this is useful when extra bandwidth is desired. A 2.4-GHz band can be used for regular network traffic and browsing, while the higher bandwidth

5-GHz band can be used for streaming movies or gaming. Conceptually, this reduces congestion in a wireless network since different functions can